# Absorbing the Mental Model

From both roadmap videos · Month 1, Week 1, Task 5

Before writing a single line of code, both roadmap videos want you to build a **mental map** of the AI engineering world. A video description is the fastest way to do this — it is a dense, author-curated summary of everything in the video: the key idea, the chapter timestamps, the resource links, and the core advice. This document extracts and explains everything both descriptions contain.

## The Two Videos at a Glance

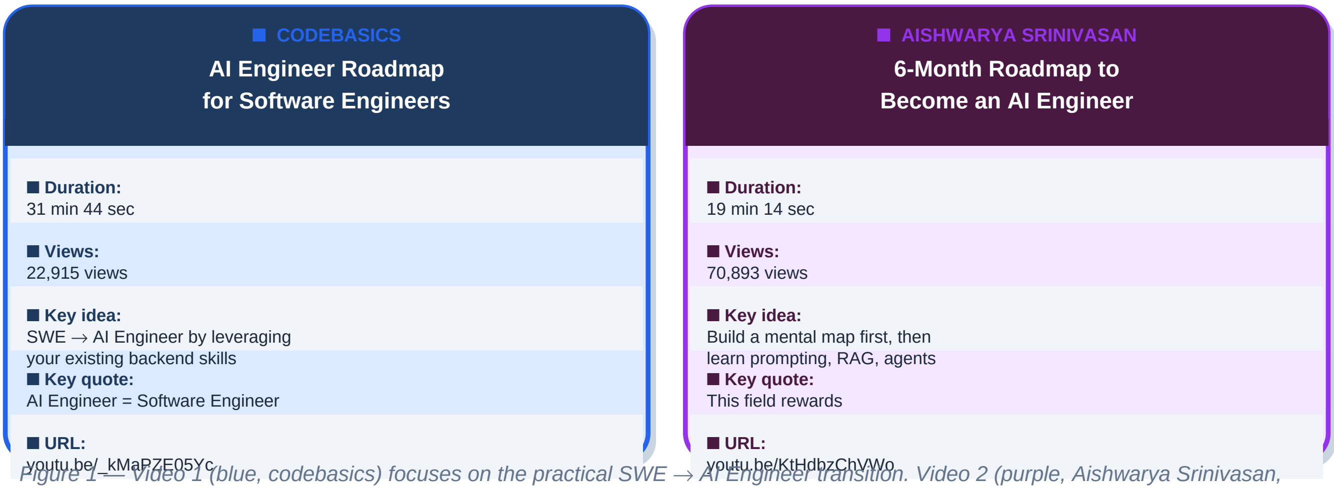


*Figure 1 — Video 1 (blue, codebasics) focuses on the practical SWE → AI Engineer transition. Video 2 (purple, Aishwarya Srinivasan, ex-Google/Microsoft) gives a 6-month structured roadmap with detailed month-by-month topics and resources.*

## Your Starting Point

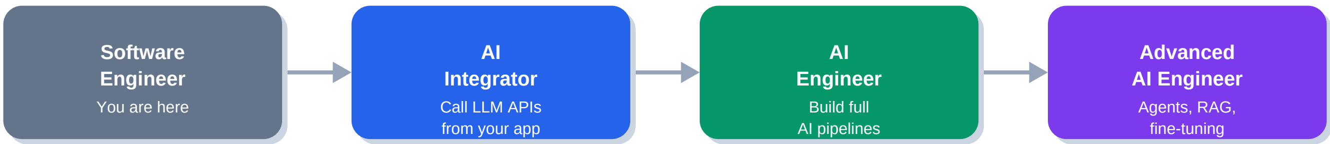


*Figure 2 — Both videos confirm the same progression. You are a Software Engineer (left). The next step is 'AI Integrator' — calling LLM APIs from your existing apps. This is already valuable and employable. Full AI Engineer and beyond come with practice.*

# Video 1 — codebasics

> ***Core definition from the description:*** *"AI Engineer = Software Engineer (backend) + LLM/AI model understanding + AI system design" — this is the most concise and accurate job description you will find anywhere.*

The description also explains *how* software engineers typically enter AI: first as **integrators** (calling LLM APIs to add AI features to existing software), then progressing to fine-tuning LLMs and eventually training ML/DL models from scratch. You do not need to jump to training models — integrating is already a well-paid, in-demand skill.

## What the Chapter Timestamps Tell You

The timestamps in a YouTube description are not just navigation aids — they are the author's own outline of the full learning path. Here is what each chapter section means for your roadmap:

| Section | Key Idea | Why it matters to you |
|---|---|---|
| **00:00 Intro 01:12 Overview** | Sets up why SWEs are best positioned to become AI engineers | Your 4 years of SWE experience is a genuine competitive advantage — not something to hide |
| **01:48 AI & Python Foundations** | Python basics, working with APIs, environment setup | You can move fast here — you already know how to code. Focus on the AI-specific parts |
| **11:56 RAG & LangChain** | Retrieval-Augmented Generation — giving AI access to your private data | This is the most common AI engineering pattern you will build in production |
| **17:19 Agentic AI Fundamentals** | AI that takes actions, uses tools, plans across multiple steps | Agents are the frontier skill that makes you senior. Build at least one agent project |
| **21:45 Observability & Deployment** | LangSmith, Langfuse, FastAPI, Docker — making AI apps production-ready | This is where your existing SWE deployment knowledge is invaluable |
| **24:40 Multi-Agent & Context Eng** | Multiple AI agents collaborating; context engineering = what you put in the prompt | Context engineering is the real 'prompt engineering' — an engineering discipline, not a soft skill |
| **28:48 DSPy** | Declarative AI programming — define prompts as modules, optimise automatically | DSPy is emerging fast. Knowing it puts you ahead of engineers who only know LangChain |
| **29:11 Azure/AWS** | Cloud deployment — video recommends picking one and going deep | You prefer AWS — AWS Bedrock, ECS, SageMaker. One cloud done well beats knowing two shallowly |
| **30:24 Tips** | Practical advice: build projects, share publicly, join communities | Both videos agree: building beats studying. Every project teaches what reading cannot |

# Video 2 — Aishwarya Srinivasan

> ***The single most important paragraph in this description:***
>
> *"An AI engineer builds applications powered by AI — primarily large language models. We are not training models from scratch. We are not doing research. We are taking powerful AI capabilities that already exist and turning them into products that solve real problems."*

This single paragraph eliminates the most common misconception about AI engineering: that you need a maths PhD or research background to do it. You do not. You build. You ship. You solve problems using existing AI capabilities through APIs. That is the job, and it is already in your reach.

## The 6-Month Roadmap from the Description

Aishwarya lays out her entire roadmap directly in the video description — month by month. This is the clearest published roadmap available. Your 4-month plan is aligned with this but compressed (you are already a SWE, so the foundations are faster for you):

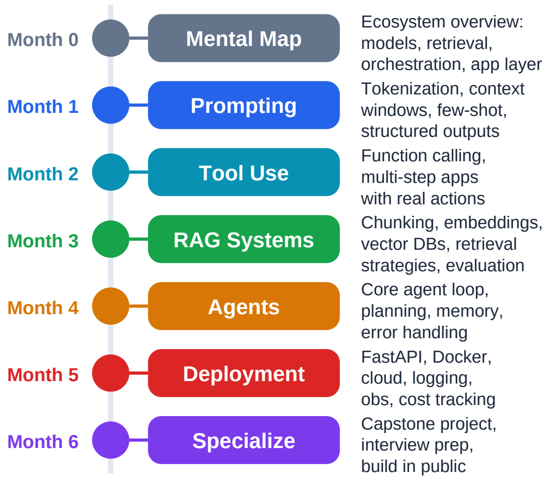


*Figure 3 — Aishwarya's 6-month roadmap, extracted directly from the video description. Month 0 is the mental map (what you are doing in Week 1). Each subsequent month builds on the last.*

> ***Closing line of the description:*** *"This field rewards builders over studiers. Every project teaches you things that reading never will. Where are you starting from?" — This applies directly to you. Your goal is to be building something by the end of Month 1.*

# The Shared Mental Model

Both videos were made independently, but they teach the same core mental model. When two experienced AI engineers from different backgrounds agree on the same structure, it means that structure reflects how the field actually works — not just one person's opinion.

## What Both Videos Agree On

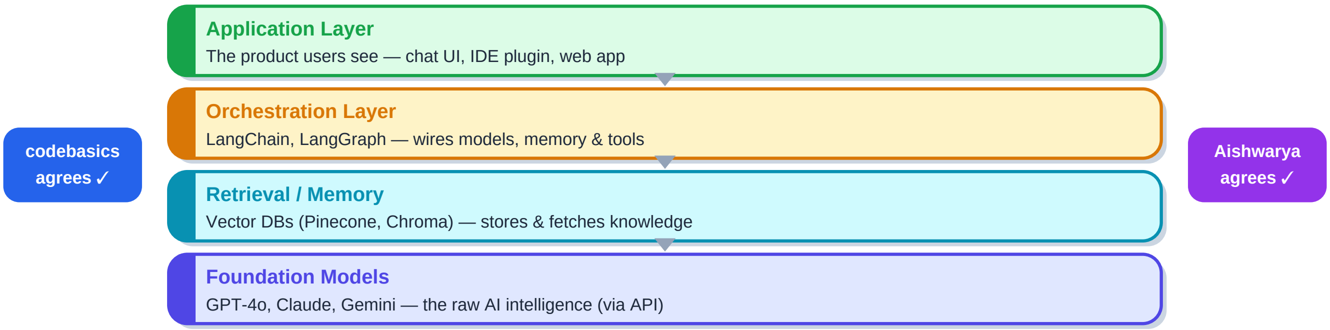


*Figure 4 — Both videos independently describe the same 4-layer architecture: Foundation Models → Retrieval/Memory → Orchestration → Application. This is your mental map. These are the four buckets every AI engineering concept fits into.*

**✓ You do not need a PhD or a research background.**
Both videos explicitly say this. Your SWE background is enough. The job is building products, not doing research.

**✓ Building beats studying.**
Aishwarya says it directly. codebasics's Tips chapter says it too. Reading PDFs and watching videos is the start — shipping projects is how you actually learn.

**✓ Start with the mental map before writing any code.**
Both videos dedicate their opening sections to ecosystem orientation — models, APIs, retrieval, orchestration, application layer. That is exactly what Week 1 of your roadmap does.

## All Resources from Both Descriptions

Both descriptions include curated resource links — these are the exact docs and tools the instructors use themselves. Bookmark these now; you will return to each one as you progress through the roadmap:

| Topic / Month | Resource | URL |
|---|---|---|
| **Month 0** | OpenAI API Docs | platform.openai.com/docs |
| **Month 0** | Anthropic API Docs | docs.anthropic.com |
| **Month 0** | Fireworks AI (model hosting) | fireworks.ai |
| **Month 0** | LangChain | langchain.com |

| Topic / Month | Resource | URL |
| --- | --- | --- |
| **Month 0** | LlamaIndex | llamaindex.ai |
| **Month 1** | OpenAI Prompt Engineering Guide | platform.openai.com/docs/guides/prompt-engineering |
| **Month 1** | Anthropic Prompt Engineering | docs.anthropic.com/en/docs/build-with-claude/prompt-engineering |
| **Month 2** | OpenAI Function Calling | platform.openai.com/docs/guides/function-calling |
| **Month 3** | Pinecone (vector DB) | pinecone.io |
| **Month 3** | Weaviate (vector DB) | weaviate.io |
| **Month 3** | ChromaDB (local vector DB) | trychroma.com |
| **Month 3** | Qdrant (vector DB) | qdrant.tech |
| **Month 4** | LangGraph (agent framework) | langchain.com/langgraph |
| **Month 5** | FastAPI | fastapi.tiangolo.com |
| **Month 5** | Docker | docs.docker.com |
| **Month 5** | LangSmith (observability) | langchain.com/langsmith |
| **Month 5** | Langfuse (open-source obs.) | langfuse.com |
| **Bonus** | codebasics Roadmap PDF | resources.codebasics.io/INV0II |

## Your Task Checklist

**Task 5 is done when all 5 items are complete:**

■ **Read Video 1 description** — Read the full description of youtube.com/watch?v=_kMaPZE05Yc. Note the formula: AI Engineer = SWE + LLM understanding + AI system design.

■ **Read Video 2 description** — Read the full description of youtube.com/watch?v=KtHdbzChVWo. Focus on the mental model paragraph and the month-by-month roadmap bullets.

■ **Write down the shared mental model** — In your own words: what do both videos agree the 4-layer structure of AI engineering is?

■ **Bookmark all resource links** — Save the 17 resource links from this page in a browser folder called 'AI Engineer Roadmap'.

■ **Identify your entry point** — Which layer of the stack are you closest to already? (Hint: Application layer — you build software.)